University of Geneva

Faculty of Science

Quantum Computing Lab 1

Selected Chapters

14x060

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2026-03-07

Contents

Exercise 1

Let $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$. Write out $|\psi\rangle^{\otimes 2}$ and $|\psi\rangle^{\otimes 3}$ explicitly, both in terms of tensor products like $|0\rangle\ |1\rangle$, and using the Kronecker product.

- As we often use the abbreviated notations $|v\rangle \otimes |w\rangle = |vw\rangle$ for the tensor product, we have in term of tensor products:

$$\begin{aligned} |\psi\rangle^{\otimes 2} &= |\psi\rangle \otimes |\psi\rangle \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (|0\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)^{1} \\ &= \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} \end{aligned}$$

- As $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$, we have by using the Kronecker[2] product:

$$\begin{aligned} |\psi\rangle^{\otimes 2} &= |\psi\rangle \otimes |\psi\rangle \\ &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \otimes \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \end{aligned}$$

- For $|\psi\rangle^{\otimes 3}$, we have in term of tensor products:

$$\begin{aligned} |\psi\rangle^{\otimes 3} &= |\psi\rangle \otimes |\psi\rangle^{\otimes 2} \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} \\ &= \frac{1}{2\sqrt{2}} (|0\rangle \otimes (|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |1\rangle \otimes (|00\rangle + |01\rangle + |10\rangle + |11\rangle)) \\ &= \frac{1}{2\sqrt{2}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle) \end{aligned}$$

[1] Distributivity of tensorial product [2] Kronecker product: Given vectors $\boldsymbol{a} \in \mathbb{C}^m$ and $\boldsymbol{b} \in \mathbb{C}^n$, their Kronecker product is defined as: $\boldsymbol{a} \otimes \boldsymbol{b} = (a_1 \boldsymbol{b}, a_2 \boldsymbol{b}, ..., a_m \boldsymbol{b})^T \in \mathbb{C}^{mn}$

- And by using the Kronecker product:

$$
\begin{aligned}
|\psi\rangle^{\otimes 3} &= |\psi\rangle \otimes |\psi\rangle^{\otimes 2} \\
&= \frac{1}{\sqrt{2}}\left(\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix}\right) \otimes \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \\
&= \frac{1}{2\sqrt{2}}\left(\begin{pmatrix}1\\1\end{pmatrix} \otimes \begin{pmatrix}1\\1\\1\\1\end{pmatrix}\right) = \begin{pmatrix}1\\1\\1\\1\\1\\1\\1\\1\end{pmatrix}
\end{aligned}
$$

Exercise 2

The Hadamard operator on one qubit may be written as

$$H = \frac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|].$$

Show explicitly that the Hadamard transform on n qubits, $H^{\otimes n}$, may be written as

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} \; |x\rangle\langle y|.$$

Write out an explicit matrix representation for $H^{\otimes 2}$.

- The notation $\langle\varphi|$ is used to represent the vector dual to $|\varphi\rangle$, so the vector φ conjugate and transposed.
- The notation $|\varphi\rangle\langle\psi|$ denote the matrix product between the vectors φ and ψ.

So we have:

$$
\begin{aligned}
H &= \frac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \\
&= \frac{1}{\sqrt{2}}\left[\begin{pmatrix}1\\1\end{pmatrix}(1 \;\; 0) + \begin{pmatrix}1\\-1\end{pmatrix}(0 \;\; 1)\right] \\
&= \frac{1}{\sqrt{2}}\left[\begin{pmatrix}1 & 0\\1 & 0\end{pmatrix} + \begin{pmatrix}0 & 1\\0 & -1\end{pmatrix}\right] \\
&= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}
\end{aligned}
$$

For $H^{\otimes 2}$, we have:

$$\begin{aligned}
H^{\otimes 2} &= H \otimes H \\
&= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & -1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}
\end{aligned}$$

We want to prove that $H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} \ |x\rangle\langle y|$ with x and y that represent n qubits, so with $|x\rangle = |x_1 x_2 ... x_n\rangle$ and $|y\rangle = |y_1 y_2 ... y_n\rangle$. So x and y are two sequences of bits.

We can define $(x \cdot y)$ as the bitwise inner product modulo 2:

$$x \cdot y = \sum_{i=1}^{n} x_i y_i \bmod 2$$

where $x_i, y_i \in \{0, 1\}$.

For the initialization, we have:

$$\begin{aligned}
H^{\otimes 1} = H &= \tfrac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \\
&= \tfrac{1}{\sqrt{2}}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|) \\
&= \tfrac{1}{\sqrt{2}}[(-1)^{0 \cdot 0} \ |0\rangle\langle 0| + (-1)^{1 \cdot 0}|1\rangle\langle 0| + (-1)^{0 \cdot 1}|0\rangle\langle 1| + (-1)^{1 \cdot 1} \ |1\rangle\langle 1|] \\
&= \tfrac{1}{\sqrt{2}} \textstyle\sum_{a,b \in \{0,1\}} (-1)^{a \cdot b} \ |a\rangle\langle b|
\end{aligned}$$

Suppose we have

$$H^{\otimes (n-1)} = \frac{1}{\sqrt{2^{n-1}}} \sum_{x', y' \in \{0,1\}^{n-1}} (-1)^{x' \cdot y'} \ |x'\rangle\langle y'|$$

By induction,

$$
\begin{aligned}
H^{\otimes n} &= H^{\otimes(n-1)} \otimes H \\
&= \frac{1}{\sqrt{2^{n-1}}} \sum_{x',y' \in \{0,1\}^{n-1}} (-1)^{x'\cdot y'} \ |x'\rangle\langle y'| \otimes \frac{1}{\sqrt{2}} \sum_{a,b\in\{0,1\}} (-1)^{a\cdot b} \ |a\rangle\langle b| \\
&= \frac{1}{\sqrt{2^{n-1}\times 2}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} (-1)^{x'\cdot y'} \ |x'\rangle\langle y'| \otimes \sum_{a,b\in\{0,1\}} (-1)^{a\cdot b} \ |a\rangle\langle b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b\in\{0,1\}} (-1)^{x'\cdot y'} (-1)^{a\cdot b} \ |x'\rangle\langle y'| \otimes |a\rangle\langle b| \right)^1 \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b\in\{0,1\}} (-1)^{x'\cdot y'} (-1)^{a\cdot b} \ |x'\rangle \otimes |a\rangle\langle y'| \otimes \langle b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b\in\{0,1\}} (-1)^{x'\cdot y'} (-1)^{a\cdot b} \ |x'a\rangle\langle y'b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x,y \in \{0,1\}^{n}} (-1)^{x\cdot y} \ |x\rangle\langle y| \right)
\end{aligned}
$$

So we have proved by induction that

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x\cdot y} \ |x\rangle\langle y|.$$

Exercise 3

(Commutation relations for the Pauli matrices) Verify the commutation relations

$$[X,Y] = 2iZ; \quad [Y,Z] = 2iX; \quad [Z,X] = 2iY.$$

There is an elegant way of writing this using ε_{jkl}, the antisymmetric tensor on three indices, for which $\varepsilon_{jkl} = 0$ except for $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = 1$, and $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$:

$$\left[\sigma_j, \sigma_k\right] = 2i \sum_{l=1}^{3} \varepsilon_{jkl}\sigma_l.$$

The commutation relation is defined as follow:

$$[A,B] = AB - BA$$

The Pauli matrix are defined as follow:

[1] We can write this thanks to the bilinearity of the tensor product: $(X_1 + X_2) \otimes (Y_1 + Y_2) = \sum_{i,j} X_i \otimes Y_j$

$$\sigma_0 \equiv I \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sigma_1 \equiv \sigma_x \equiv X \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 \equiv \sigma_y \equiv Y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 \equiv \sigma_z \equiv Z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

So we have:

- $$\begin{aligned} [X, Y] &= XY - YX \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\ &= \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} \\ &= 2i \times \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= 2iZ \end{aligned}$$
- $$\begin{aligned} [Y, Z] &= YZ - ZY \\ &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix} \\ &= 2i \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= 2iX \end{aligned}$$

- $$\begin{aligned}[Z,X] &= ZX - XZ \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} \\ &= 2 \times \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &= 2 \times \begin{pmatrix} 0 & (-i)^2 \\ i^2 & 0 \end{pmatrix} \\ &= 2iY\end{aligned}$$

Exercise 4

Suppose $[A, B] = 0$, $\{A, B\} = 0$, and A is invertible. Show that B must be 0.

The anticommutation relation is defined as follow:

$$\{A, B\} = AB + BA$$

So if we have $[A, B] = 0$, $\{A, B\} = 0$, and A invertible, it means that

$$\begin{aligned} &[A, B] = \{A, B\} \\ \Leftrightarrow\ &AB - BA = AB + BA \\ \Leftrightarrow\ &2BA = 0 \\ \Leftrightarrow\ &BAA^{-1} = 0A^{-1} \\ \Leftrightarrow\ &B = 0 \end{aligned}$$

So B must be 0.

Exercise 5

Suppose A and B are commuting Hermitian operators. Prove that $\exp(A)\exp(B) = \exp(A + B)$.

An operator is Hermitian if its transpose conjugate is equal to the operator itself.

As A and B are commuting, we have $[A, B] = 0 \Leftrightarrow AB - BA = 0 \Leftrightarrow AB = BA$.

The Taylor expansion of the exponential function for an operator X is:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

So we have:

$$\begin{aligned}
\exp(A)\exp(B) &= \sum_{k=0}^{\infty} \frac{A^k}{k!} \sum_{j=0}^{\infty} \frac{B^j}{j!} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{A^k B^j}{k! j!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{A^k B^{n-k}}{k!(n-k)!} && \text{let } \ n = j + k \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{n!}{n!k!(n-k)!} \left(A^k B^{n-k}\right) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \left(A^k B^{n-k}\right) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} \left(A^k B^{n-k}\right) && \text{by definition of } \binom{n}{k} = \frac{n!}{k!(n-k)!} \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} (A+B)^n && \text{according to binomial theorem}^{1} \\
&= \exp(A+B) && \text{according to taylor expansion}
\end{aligned}$$

Exercise 6

Suppose we have a qubit in the state $|0\rangle$, and we measure the observable X. What is the average value of X? What is the standard deviation of X?

The state of a qubit can be described as the combination of two states: $|0\rangle$ and $|1\rangle$. So the state of a qubit $|\varphi\rangle$ can be written as

$$|\varphi\rangle = \alpha\ |0\rangle + \beta\ |1\rangle$$

with $\alpha, \beta \in \mathbb{C}$.

As the qubit state is a combination of two states, the qubit state cannot be observed. However, the qubit state can be measured to fix the state of the qubit to state $|0\rangle$ or state $|1\rangle$.

According to the Born Rule, for a qubit $|\varphi\rangle = \alpha\ |0\rangle + \beta\ |1\rangle$, the probability that the qubit is in the state $|0\rangle$ is given by $|\alpha|^2$ and the probability that the qubit is in the state $|1\rangle$ is given by $|\beta|^2$. As $|\alpha|^2$ and $|\beta|^2$ are probabilities, we have that $|\alpha|^2 + |\beta|^2 = 1$.

If we consider that we have a qubit in the state $|0\rangle$, it means that $|\alpha|^2 = 1$ and $|\beta|^2 = 0$ since $|\alpha|^2 + |\beta|^2 = 1 \Leftrightarrow 1 + |\beta|^2 = 1 \Leftrightarrow |\beta|^2 = 0 \Leftrightarrow \beta = 0$. As α is a complex number, we cannot determine the value of α.

So the average value of the observable X is given by:

[1] Binomial theorem: $(A+B)^n = \sum_{k=0}^{n} \binom{n}{k} A^k B^{n-k}$

$$\mathbb{E}[X] = \alpha\ |0\rangle + \beta\ |1\rangle = \alpha\ |0\rangle$$

The standard deviation of X is given by:

$$\begin{aligned}
\mathrm{Var}[X] &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\
&= \mathbb{E}\big[(\alpha\ |0\rangle + \beta\ |1\rangle)^2\big] - (\alpha\ |0\rangle)^2 \\
&= \mathbb{E}\left[\alpha^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T\right] - \alpha^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \\
&= \alpha^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T - \alpha^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \\
&= 0
\end{aligned}$$

Exercise 7

Calculate the probability of obtaining the result +1 for a measurement of $\vec{v} \cdot \vec{\sigma}$, given that the state prior to measurement is $|0\rangle$. What is the state of the system after the measurement if +1 is obtained?

By definition, we have our measurement $M = \vec{v} \cdot \vec{\sigma} = v_1\sigma_X + v_2\sigma_Y + v_3\sigma_Z$ with $\|\vec{v}\|^2 = 1$.

So:

$$\begin{aligned}
M &= v_1\sigma_X + v_2\sigma_Y + v_3\sigma_Z \\
&= v_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + v_2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + v_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} 0 & v_1 \\ v_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -iv_2 \\ iv_2 & 0 \end{pmatrix} + \begin{pmatrix} v_3 & 0 \\ 0 & -v_3 \end{pmatrix} \\
&= \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}
\end{aligned}$$

The observable M has a spectral decomposition given by

$$M = \sum_m m P_m$$

where P_m is the projector onto the eigenspace of M with eigenvalue m.

The projector P_m satisfy the relation $\sum_m P_m = I$.

The state of the quantum system after measurement is given by the formula

$$\frac{P_m|\varphi\rangle}{\sqrt{p(m)}}$$

with $p(m) = \langle\varphi \mid P_m \mid \varphi\rangle$ the probability to obtain result m.

We want to find eigenvalues of M, so we want to find λ such that $\det(M - \lambda I) = 0$. We have:

$$\begin{aligned}
\det(M-\lambda I) &= \det\left(\begin{pmatrix} v_3-\lambda & v_1-iv_2 \\ v_1+iv_2 & -v_3-\lambda \end{pmatrix}\right) \\
&= (v_3-\lambda)(-v_3-\lambda)-(v_1-iv_2)(v_1+iv_2) \\
&= (-\lambda+v_3)(-\lambda-v_3)-(v_1-iv_2)(v_1+iv_2) \\
&= \lambda^2-(v_3^2+v_1^2+v_2^2) \\
&= \lambda^2-\|\vec{v}\|^2 \\
&= \lambda^2-1
\end{aligned}$$

So $\det(M-\lambda I)=0 \iff \lambda^2-1=0 \iff \lambda=\pm 1$

So we have: $M=P_1-P_{-1}$.

We know that the result $m=+1$ is obtained.

So we want to find value of projector P_1 that correspond to eigenvalue $+1$.

As $M=P_1-P_{-1}$ and $\sum_m P_m = I$, we have:

$$I+M = P_1+P_{-1}+P_1-P_{-1} = 2P_1 \iff P_1 = \frac{I+M}{2}$$

So

$$\begin{aligned}
P_1 &= \frac{1}{2}(I+M) \\
&= \frac{1}{2}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} v_3 & v_1-iv_2 \\ v_1+iv_2 & -v_3 \end{pmatrix}\right) \\
&= \frac{1}{2}\begin{pmatrix} v_3+1 & v_1-iv_2 \\ v_1+iv_2 & -v_3+1 \end{pmatrix}
\end{aligned}$$

Now, we can compute the probability $p(+1)$. We know that the state prior to measurement is $\varphi = |0\rangle$.

$$\begin{aligned}
p(+1) &= \langle 0 \mid P_1 \mid 0 \rangle \\
&= \frac{1}{2}(1\ \ 0)\begin{pmatrix} v_3+1 & v_1-iv_2 \\ v_1+iv_2 & -v_3+1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \frac{1}{2}(v_3+1\ \ v_1-iv_2)\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \frac{v_3+1}{2}
\end{aligned}$$

So the state of the quantum system after measurement is:

$$\begin{aligned}
\frac{P_m|\varphi\rangle}{\sqrt{p(m)}} &= \frac{P_1|0\rangle}{\sqrt{p(+1)}} \\
&= \frac{1}{2}\frac{1}{\sqrt{(v_3+1)/2}}\begin{pmatrix} v_3+1 & v_1-iv_2 \\ v_1+iv_2 & -v_3+1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \frac{1}{\sqrt{2(v_3+1)}}\begin{pmatrix} v_3+1 \\ v_1+iv_2 \end{pmatrix}
\end{aligned}$$

Exercise 8

Show that the average value of the observable $X_1 Z_2$ for a two qubit system measured in the state $(|00\rangle + |11\rangle)/\sqrt{2}$ is zero.

First, $X_1 Z_2$ means that the Pauli matrix X is applied to qubit 1 and Pauli matrix Z is applied to qubit 2. So it means that $X_1 Z_2 = X \otimes Z$. Indeed, $(X \otimes Z)|ab\rangle = (X|a\rangle) \otimes (Z|b\rangle)$.

First of all, we need to study impact of Pauli matrices on single qubit. We have:

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$
$$X|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$
$$Z|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$
$$Z|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -|1\rangle$$

So the X Pauli matrix transform state $|1\rangle$ to state $|0\rangle$ and state $|0\rangle$ to state $|1\rangle$, and the Z Pauli matrix transform state $|1\rangle$ to state $-|1\rangle$ and leaves the state $|0\rangle$ unchanged.

We have:

$$\langle ab|cd\rangle = (\langle a| \otimes \langle b|)(|c\rangle \otimes |d\rangle) = \langle a|c\rangle \otimes \langle b|d\rangle$$

and:

$$\langle 0|0\rangle = (1 \;\; 0) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$$
$$\langle 0|1\rangle = (1 \;\; 0) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$
$$\langle 1|0\rangle = (0 \;\; 1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$$
$$\langle 1|1\rangle = (0 \;\; 1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1$$

So as the tensor product between two scalar is a simple multiplication, we have:

$$\langle ab|cd\rangle = \begin{cases} 1 \text{ if } a = c \text{ and } b = d \\ 0 \text{ else} \end{cases}$$

The average value of an observable M for a state $|\psi\rangle$ is given by: $\langle \psi \mid M \mid \psi \rangle$.

We have $|\psi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and $M = X_1 Z_2 = X \otimes Z$.

So the average value of the observable $X_1 Z_2$ is:

$$
\begin{aligned}
\langle \psi \mid M \mid \psi \rangle &= \frac{\langle 00| + \langle 11|}{\sqrt{2}} X_1 Z_2 \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\
&= \frac{1}{2}(\langle 00| + \langle 11|)(X_1 Z_2 |00\rangle + X_1 Z_2 |11\rangle) \\
&= \frac{1}{2}(\langle 00| + \langle 11|)(|10\rangle - |01\rangle) \\
&= \frac{1}{2}(\langle 00|10\rangle + \langle 11|10\rangle - \langle 00|01\rangle - \langle 11|01\rangle) \\
&= \frac{1}{2}(0 + 0 - 0 - 0) \\
&= 0
\end{aligned}
$$

Exercise 9

Verify that the Bell basis forms an orthonormal basis for the two qubit state space.

The Bell basis is given by:

$$
\begin{aligned}
00 &: \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |e_0\rangle \\
01 &: \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |e_1\rangle \\
10 &: \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |e_2\rangle \\
11 &: \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |e_3\rangle
\end{aligned}
$$

To verify that the Bell basis forms an orthonormal basis for the two qubit state space, we need to check if each element is orthogonal with the others and if each element has a norm of 1.

Orthogonality

We will use the result of the previous exercise:

$$
\langle ab|cd\rangle = \begin{cases} 1 \text{ if } a = c \text{ and } b = d \\ 0 \text{ else} \end{cases}
$$

So we have:

$$\langle e_0 \mid e_1\rangle = \frac{\langle 00| + \langle 11|}{\sqrt{2}} \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|00\rangle - \langle 00|11\rangle + \langle 11|00\rangle - \langle 11|11\rangle) = \frac{1}{2}(1-1) = 0$$

$$\langle e_0 \mid e_2\rangle = \frac{\langle 00| + \langle 11|}{\sqrt{2}} \frac{|10\rangle + |01\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|10\rangle + \langle 00|01\rangle + \langle 11|10\rangle + \langle 11|01\rangle) = 0$$

$$\langle e_0 \mid e_3\rangle = \frac{\langle 00| + \langle 11|}{\sqrt{2}} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|01\rangle - \langle 00|10\rangle + \langle 11|01\rangle - \langle 11|10\rangle) = 0$$

$$\langle e_1 \mid e_2\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \frac{|10\rangle + |01\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|10\rangle + \langle 00|01\rangle - \langle 11|10\rangle - \langle 11|01\rangle) = 0$$

$$\langle e_1 \mid e_3\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|01\rangle - \langle 00|10\rangle - \langle 11|01\rangle + \langle 11|10\rangle) = 0$$

$$\langle e_2 \mid e_3\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 10|01\rangle - \langle 10|10\rangle + \langle 01|01\rangle + \langle 01|10\rangle) = \frac{1}{2}(-1+1) = 0$$

So the Bell basis form an orthogonal basis for the two qubit state space.

Unitary norm

The norm of a vector is defined as the square root of its inner product with itself. So we have:

$$\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle}$$

So:

$$\langle e_0|e_0\rangle = \frac{\langle 00| + \langle 11|}{\sqrt{2}} \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|00\rangle + \langle 00|11\rangle + \langle 11|00\rangle + \langle 11|11\rangle) = \frac{1}{2}(1+1) = 1$$

$$\langle e_1|e_1\rangle = \frac{\langle 00| - \langle 11|}{\sqrt{2}} \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 00|00\rangle - \langle 00|11\rangle - \langle 11|00\rangle + \langle 11|11\rangle) = \frac{1}{2}(1+1) = 1$$

$$\langle e_2|e_2\rangle = \frac{\langle 10| + \langle 01|}{\sqrt{2}} \frac{|10\rangle + |01\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 10|10\rangle + \langle 10|01\rangle + \langle 01|10\rangle + \langle 01|01\rangle) = \frac{1}{2}(1+1) = 1$$

$$\langle e_3|e_3\rangle = \frac{\langle 01| - \langle 10|}{\sqrt{2}} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{2}(\langle 01|01\rangle - \langle 01|10\rangle - \langle 10|01\rangle + \langle 10|10\rangle) = \frac{1}{2}(1+1) = 1$$

So as each element of the Bell basis is orthogonal with the others and have a norm of 1, the Bell basis is an orthonormal basis.

Exercise 10

Suppose E is any positive operator acting on Alice's qubit. Show that $\langle\psi|E \otimes I|\psi\rangle$ *takes the same value* when $|\psi\rangle$ is any of the four Bell states. Suppose some malevolent third party ('Eve') intercepts Alice's qubit on the way to Bob in the superdense coding protocol. Can Eve infer anything about which of the four possible bit strings $00, 01, 10, 11$ Alice is trying to send? If so, how, or if not, why not?

The Bell basis is given by:

$$
\begin{aligned}
00 &: \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |e_0\rangle \\
01 &: \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |e_1\rangle \\
10 &: \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |e_2\rangle \\
11 &: \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |e_3\rangle
\end{aligned}
$$

As proved in exercise 9, the Bell basis is an orthonormal basis.

So $\operatorname{tr}(|e_i\rangle\langle e_i|) = \langle e_i|e_i\rangle = 1$

On top of that, we have from page 76 of supplement material:

$$\operatorname{tr}(A|\psi\rangle\langle\psi|) = \sum_i \langle i|A|\psi\rangle\langle\psi|i\rangle = \langle\psi|A|\psi\rangle$$

and from page 107:

$$\operatorname{tr}(M \operatorname{tr}_B(\rho_{AB})) = \operatorname{tr}((M \otimes I)\rho_{AB})$$

with tr_B the partial trace over system B, defined as follow:

$$\operatorname{tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = \operatorname{tr}_B(|a_1 b_1\rangle\langle a_2 b_2|) = |a_1\rangle\langle a_2| \operatorname{tr}(|b_1\rangle\langle b_2|)$$

As we consider that $|\psi\rangle$ takes any of the four Bell states, we have:

$$
\begin{aligned}
\operatorname{tr}_B(|e_0\rangle\langle e_0|) &= \operatorname{tr}_B\left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \frac{\langle 00| + \langle 11|}{\sqrt{2}}\right) \\
&= \operatorname{tr}_B\left(\frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)\right) \\
&= \frac{1}{2}(\operatorname{tr}_B(|00\rangle\langle 00|) + \operatorname{tr}_B(|00\rangle\langle 11|) + \operatorname{tr}_B(|11\rangle\langle 00|) + \operatorname{tr}_B(|11\rangle\langle 11|)) \\
&= \frac{1}{2}(|0\rangle\langle 0| \operatorname{tr}(|0\rangle\langle 0|) + |0\rangle\langle 1| \operatorname{tr}(|0\rangle\langle 1|) + |1\rangle\langle 0| \operatorname{tr}(|1\rangle\langle 0|) + |1\rangle\langle 1| \operatorname{tr}(|1\rangle\langle 1|)) \\
&= \frac{1}{2}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right) + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \operatorname{tr}\left(\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right)\right. \\
&\quad \left. + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \operatorname{tr}\left(\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \operatorname{tr}\left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right)\right) \\
&= \frac{1}{2}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) \\
&= \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
&= \frac{1}{2} I
\end{aligned}
$$

$$
\begin{aligned}
\operatorname{tr}_B(|e_1\rangle\langle e_1|) &= \operatorname{tr}_B\left(\frac{|00\rangle - |11\rangle}{\sqrt{2}}\frac{\langle 00| - \langle 11|}{\sqrt{2}}\right) \\
&= \operatorname{tr}_B\left(\frac{1}{2}(|00\rangle\langle 00| - |00\rangle\langle 11| - |11\rangle\langle 00| + |11\rangle\langle 11|)\right) \\
&= \frac{1}{2}(\operatorname{tr}_B(|00\rangle\langle 00|) - \operatorname{tr}_B(|00\rangle\langle 11|) - \operatorname{tr}_B(|11\rangle\langle 00|) + \operatorname{tr}_B(|11\rangle\langle 11|)) \\
&= \frac{1}{2}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) \\
&= \frac{1}{2}I
\end{aligned}
$$

$$
\begin{aligned}
\operatorname{tr}_B(|e_2\rangle\langle e_2|) &= \operatorname{tr}_B\left(\frac{|10\rangle + |01\rangle}{\sqrt{2}}\frac{\langle 10| + \langle 01|}{\sqrt{2}}\right) \\
&= \operatorname{tr}_B\left(\frac{1}{2}(|10\rangle\langle 10| + |10\rangle\langle 01| + |01\rangle\langle 10| + |01\rangle\langle 01|)\right) \\
&= \frac{1}{2}(\operatorname{tr}_B(|10\rangle\langle 10|) + \operatorname{tr}_B(|10\rangle\langle 01|) + \operatorname{tr}_B(|01\rangle\langle 10|) + \operatorname{tr}_B(|01\rangle\langle 01|)) \\
&= \frac{1}{2}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) \\
&= \frac{1}{2}I
\end{aligned}
$$

$$
\begin{aligned}
\operatorname{tr}_B(|e_3\rangle\langle e_3|) &= \operatorname{tr}_B\left(\frac{|01\rangle - |10\rangle}{\sqrt{2}}\frac{\langle 01| - \langle 10|}{\sqrt{2}}\right) \\
&= \operatorname{tr}_B\left(\frac{1}{2}(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|)\right) \\
&= \frac{1}{2}(\operatorname{tr}_B(|01\rangle\langle 01|) - \operatorname{tr}_B(|01\rangle\langle 10|) - \operatorname{tr}_B(|10\rangle\langle 01|) + \operatorname{tr}_B(|10\rangle\langle 10|)) \\
&= \frac{1}{2}\left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) \\
&= \frac{1}{2}I
\end{aligned}
$$

So we have $\operatorname{tr}_B(|\psi\rangle\langle\psi|) = \frac{1}{2}I$ for any of the four Bell states $|\psi\rangle$.

So we have:

$$
\begin{aligned}
\langle\psi|(E \otimes I)|\psi\rangle &= \operatorname{tr}((E \otimes I)|\psi\rangle\langle\psi|) \\
&= \operatorname{tr}(E \operatorname{tr}_B(|\psi\rangle\langle\psi|)) \\
&= \operatorname{tr}\left(\frac{EI}{2}\right) \\
&= \frac{1}{2}\operatorname{tr}(E)
\end{aligned}
$$

So $\langle\psi|E \otimes I|\psi\rangle$ takes the same value $\frac{1}{2}\operatorname{tr}(E)$ when $|\psi\rangle$ is any of the four Bell states.

For superdense coding protocol, Alice and Bob share a pair of qubits in the entangled state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$. So $|\psi\rangle$ cannot be written as a product of two single qubit states.

Alice want to send two classical bits of information to Bob.

As describen in page 97-98 of the supplement material, Alice can apply a unitary operation on her qubit to modify the global state of the two qubits. To apply any unitary operation on her qubit and leave Bob's qubit unchanged, she applies $E \otimes I$ with E the unitary operation she wants to apply.

Indeed, thanks to the property of the tensor product, we have: $|ab\rangle(E \otimes I) = (E|a\rangle) \otimes (I|b\rangle)$ which modifies only the state of Alice's qubit.

So if she wishes to send the bit string 00, she does nothing to her qubit. If she wishes to send the bit string 01, she applies the Pauli matrix Z to her qubit. If she wishes to send the bit string 10, she applies the Pauli matrix X to her qubit. If she wishes to send the bit string 11, she applies the Pauli matrix iY to her qubit.

The four resulting states obtained are the four Bell states:

$$00 : \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |e_0\rangle$$

$$01 : \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |e_1\rangle$$

$$10 : \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |e_2\rangle$$

$$11 : \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |e_3\rangle$$

So $|\psi\rangle$ is any of the four Bell states.

Bob doesn't know which of the four Bell states $|\psi\rangle$ is. But as the Bell basis is orthonormal, he can check which state $|\psi\rangle$ it is by performing the inner product of $|\psi\rangle$ with each of the four Bell states. The result will be 1 for the Bell state that is equivalent to the Bell state $|\psi\rangle$.

If Eve intercepts the qubit sent by Alice, she must measure: $\langle\psi \mid (E \otimes I) \mid \psi\rangle$.

As $\langle\psi|E \otimes I|\psi\rangle$ takes the same value $\frac{1}{2}\,\mathrm{tr}(E)$ for any of the four Bell states, no informations are send about the bit string Alice is trying to send. Indeed, Eve doesn't know which of the four Bell states Alice has created.

So Eve cannot infer anything about which of the four possible bit strings $00, 01, 10, 11$ Alice is trying to send.